

Problem Set #3: Classical Mechanics (PH31207)

August 31, 2026

Virtual Work and D'Alembert's Principle

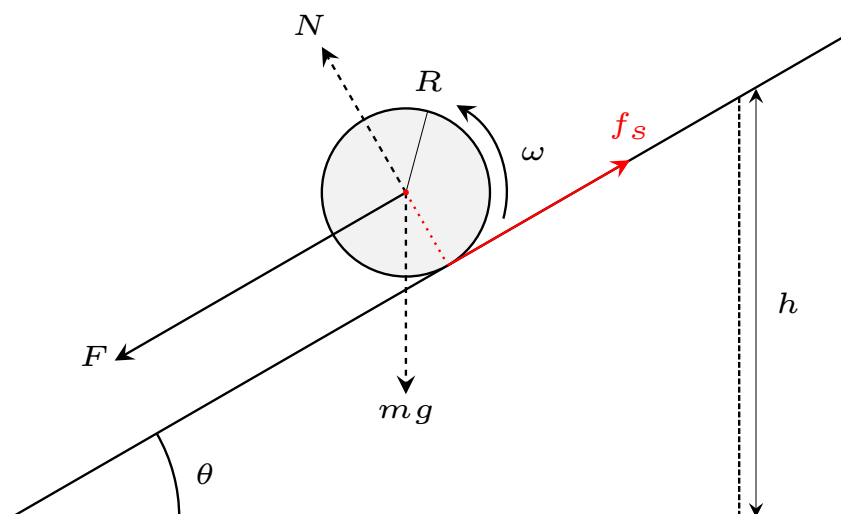


Figure 1: A hoop rolling down a fixed inclined plane.

1. A hoop of mass m is initially at rest on a smooth frictionless inclined plane at a vertical height h above the ground. Assuming the hoop slides without friction, determine its acceleration.
2. A hoop of mass m and radius R rolls without slipping down a fixed inclined plane inclined at an angle θ to the horizontal as shown in Fig. 1.
 - (a) Using D'Alembert's principle, obtain the equation of motion and determine the acceleration of the hoop. Compare this acceleration with the frictionless sliding case.

- (b) Now solve the same problem using the Lagrange's equations of the first kind, and calculate the constraint force due to friction.

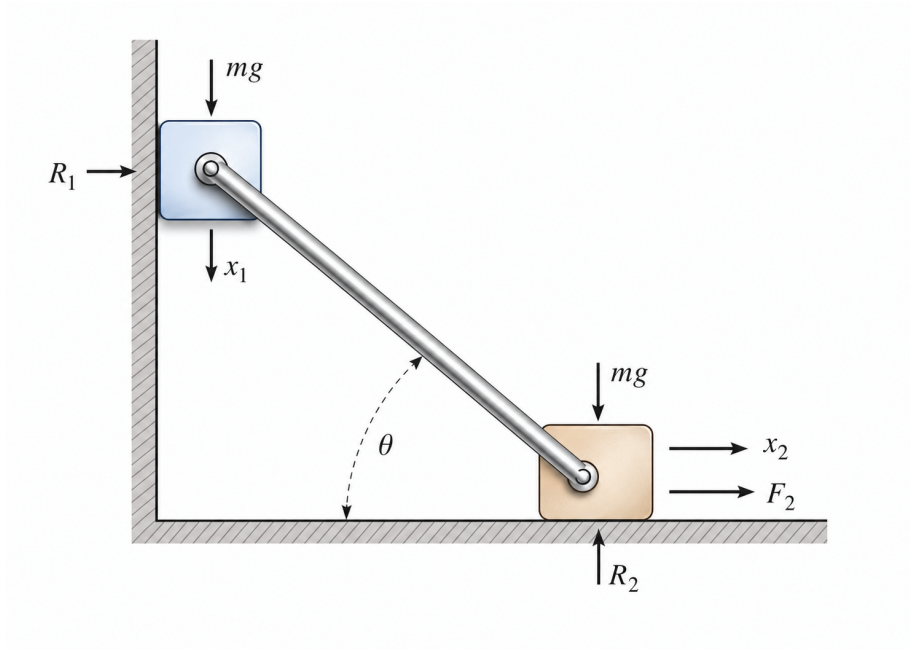


Figure 2: Taken from “Classical Dynamics” by D. T. Greenwood, pp. 19 – 20.

- In Fig. 2, the applied forces are the gravitational forces acting on the blocks and the external force F_2 . Calculate the force required $F_2(\theta)$ to keep the initially motionless system in static equilibrium.
- The purpose of this problem is to relate Q with P for a particular configuration (φ, ψ, x) at equilibrium using the principle of virtual work

$$\sum_i \vec{F}_i^{(a)} \cdot \delta \vec{r}_i \Big|_{\text{at equilibrium}} = 0.$$

In Fig. 3, P is the total force due to steam pressure on the piston, and Q is the applied opposing force.

- How many degrees of freedom does this system have?
- Write down the constraint relations.
- Using these two relations relate δx to $\delta \varphi$.
- Finally, relate Q with P .

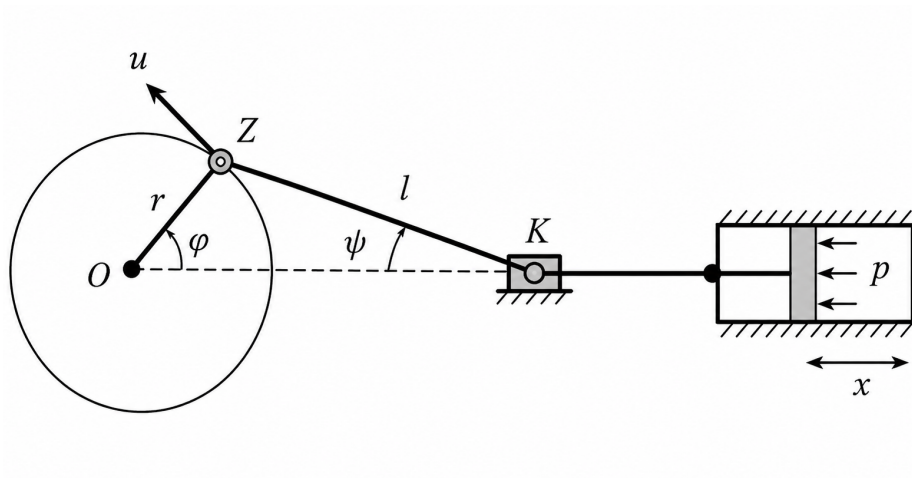


Figure 3: Taken from “Mechanics, Lectures on Theoretical Physics, Vol. 1” by A. Sommerfeld, pp. 51 – 58. The steam pressure acting on the piston is transmitted by the piston rod to the crosshead K . The connecting rod acts on the crank pin Z with a thrust which has the direction of the rod. The component in the direction of the crank – i.e., toward the center of the crank shaft – is absorbed by the rigidly fixed shaft bearing: O . In order that the system is in equilibrium, only that part of the thrust U , which is perpendicular to the crank, and therefore tangential to the crank circle, need be opposed by an equal applied force Q .